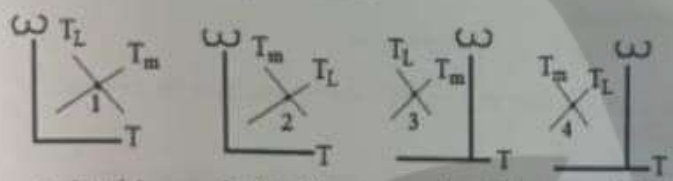
Al-Balqa' Applied University
Faculty Of Engineering Technology
Mechatronics Engineering Dept.

Time of Exam: 80 Min.
12/09/2022

Fundamentals of Elec. Drive	**Final exam 50 Points**	Prof. Dr. Mahdi Alshamasin / **Instructor**
الرقم الجامعي:	الشعبة:	اسم الطالب

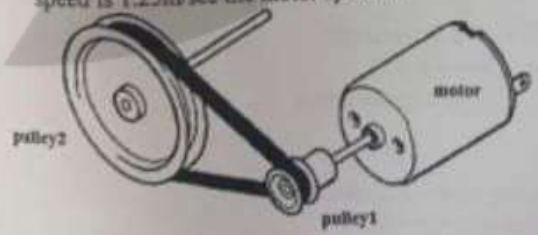
Q1: (35 marks)
Answer the following questions by filling the table at the end of this question

1. Unstable points of equilibrium are



a. 1 and 4 b. 2 and 3 c. 1 and 3 d. 2 and 4

2. Conveyor torque –speed characteristic is:
   a. Constant torque characteristic b. Linearly rising torque characteristic
   c. Non-Linearly rising torque characteristic d. Non-Linearly decreasing torque characteristic
3. If the torque developed by the motor is $T = T_L \mp J\frac{d\omega}{dt}$, the motor is working at
   a. no-load condition b. transient condition c. S.S.P d. none of these
4. The motion equation $T = T_L + J\frac{d\omega}{dt} + B\omega$ can be solved analytically if
   a. T and T_L are constants b. T or T_L is neither constant nor proportional to speed
   c. T and T_L are constants or proportional to speed d. none of these.
5. The equivalent moment of inertia referred to motor shaft calculated depending on:
   a. Heating (electric) power conservation law b. Mechanical power conservation law
   c. Electro-magnetic energy conservation law d. Kinetic energy conservation law
6. In the figure below, diameters of pulley 1 and pulley2 are 10cm and 50cm respectively, if the belt speed is 1.25m/sec the motor speed is



a.240rpm b. 250rpm c. 12r/sec d. none of these

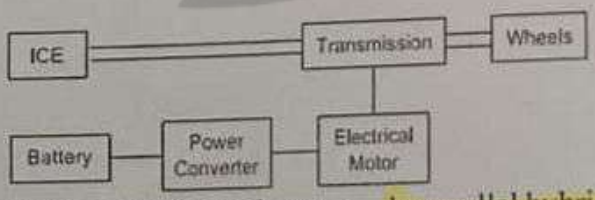
7. One of the following dc motors does not operate in regenerating mode:
   a. Parallel field dc motor b. Series field dc motor
   c. Compound field dc motor d. Separately excited dc motor
8. When the motor speed is controlling by stator voltage changing, the permissible torque is governed by
   a. $T_{per} = T_n S/S_n$ b. $T_{per} = T_n S_{sc}/S$ c. $T_{per} = T_n S_n/S$ d. $T_{per} = T_{cr}S_{cr}/S$
9. Component determines the motor starting time is
   a. motor inertia b. motor load c. speed change rate d. all of these

15. In electric motors, heating time constant is
a. equal to the electrical time constant b. more than the electrical and mechanical time constants
c. less than the electrical and mechanical time constants d. equal to the mechanical time constant
16. In the short time duty, the motor nominal torque is
a. equal to the load torque b. more than the load torque
c. less than the load torque d. more or equal to the load torque
17. In the intermittent periodic duty, the motor temperature can
a. reach the ambient temperature at rest period b. reach the maximum temperature at first period
c. fluctuate between the maximum temperature and the ambient temperature
d. reach the maximum temperature at many periods
18.In squirrel-cage motor,
a. there is a direct access to the rotor quantities such as rotor fluxes and currents
b. there is no direct access to the stator quantities such as stator fluxes and currents
c. control of decoupled torque and flux producing components of the stator phase currents is easy
d. there is a direct access to the rotor quantities such as rotor angular speed and position
19. In closed-loop vector control with an encoder to provide shaft position feedback,
a. the controller mustn't be tuned for proper operation b. the controller must be tuned for proper operation
c. it's important to have a very accurate mathematical model of the motor d. none of these
20.In Park transformation module, the following matrix is true

a. $\begin{bmatrix} V_{s\alpha} \\ V_{s\beta} \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} V_{sd} \\ V_{sq} \end{bmatrix}$ b. $\begin{bmatrix} i_{sd} \\ i_{sq} \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} i_{s\alpha} \\ i_{s\beta} \end{bmatrix}$

c. $\begin{bmatrix} i_{s\alpha} \\ i_{s\beta} \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 1 & -1/2 & -1/2 \\ 0 & \sqrt{3}/2 & -\sqrt{3}/2 \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix}$ d. none of these

21. For energy saving in drive systems, it's useful use
a. resistance controllers b. eddy-current Couplings
c. semiconductor converters d. Ward -Leonard set
22. Power factor of electric drives can be improved by using of
a. Inductors b. resistors c. L-R circuits d. capacitors
23. In pump drives where fluid flow is controlled, operation which could lead to saving of substantial amount of energy is
a. application of variable speed drives b. adjustment of opening of valves (by throttling)
c. adjustment by passing excess output d. all of these
24. The following block diagram represents



a. series hybrid architecture b. parallel hybrid architecture c. series-parallel hybrid architecture
d. pure electric vehicle
25. The stator waveform of the permanent magnet synchronous motor (PMSM) will be
a. trapezoidal b. rectangle c. sinusoidal d. square
26. A flux weakening mode that enables a higher speed operation in the constant power region is possible in PMSM by
a. applying a stator flux in opposition to the rotor magnet flux
b. increasing the stator frequency above the rated value
c. decreasing the stator voltage down the rated value
d. none of these
27. An 400V, 50Hz induction motor was used to drive production machinery, if the stator frequency is changed to 35Hz, the stator voltage should be changed to

10. Component determines the motor starting time is
a. motor inertia b. motor load c. speed change rate d. all of these

11. The following expression $\frac{3V_1^2 R_2'}{\omega_s\left(\left(R_1+R_2'\right)^2+\left(X_{SC}\right)^2\right)}$ determines:
a. Maximum Torque b. Motor no-load torque
c. Starting torque d. Motoring mode torque

12. If the variable voltage variable frequency method is used and the load is ventilator, the low which governs the motor operation is
a. $\frac{V}{\sqrt{f}}$=constant b. $\frac{V}{f^2}$=constant c. $\frac{V}{f}$=constant d. none of these

13. If the line voltage and frequency are V_L=380V, f=50Hz, then the most suitable motor to be started by Y/Δ is:
a. Y/Δ 660/1125V b. Y/Δ 660/380V
c. Y/Δ 380/220V d. Y/Δ 220/110V

14. Braking mode that is used to store reactive energy:
a. dynamic b. zero-sequence braking c. plugging d. regenerative

15. A Y- connected three-phase motor consumes 10A from the line, if it is reconnected by Δ for the same line voltage it will consume current equals to
a. 30A b. 10A c. 17A d. 50A

16. If the synchronous motor load torque value is more than synchronizing torque during starting, the motor will:
a. Run in short circuit mode
b. Not be synchronized and operate as asynchronous motor
c. Be synchronized and operate as synchronous motor
d. Be synchronized and operate as synchronous generator

17. If the nominal speed of squirrel- cage motor is 980 rpm, the motor has
a. 2 poles b. 4 poles c. 6 poles d. none of these

$$P_{eq}=\sqrt{\frac{\sum_{i=1}^{n} P_i^2 t_i}{\sum_{i=1}^{n} t}}$$

18. The relationship used to calculate equivalent load power of:
a. Continuous duty cycle with constant load b. Continuous duty cycle with fluctuating periodic loading
c. Short time duty cycle d. Continuous duty cycle with intermittent periodic loading

19. Indirect starting of electric motor is used when:
a. The motor is medium and high power and the source is limited power supply
b. The motor is fractional power and the source is unlimited power supply
c. The motor is medium and high power and the source is unlimited power supply
d. none of these

20. The dc motors allowed starting current value is:
a. (2-3) times of nominal current b. (5-7) times of nominal current
c. Nominal current d. Less than nominal current

21. The ac motor power rating is
a. proportional to the motor nominal slip b. not depending on the motor nominal slip
c. inversely proportional to the motor nominal slip d. none of these

22. In the intermittent periodic duty, the motor temperature can
a. reach the equilibrium temperature at rest period
b. reach the equilibrium temperature at first period

Q3: (7 marks)

An induction motor has a stator resistance of 3Ω and a rotor resistance referred to the stator is 2Ω, the critical inductive reactance X_{cr}=10Ω, calculate

a. The change in starting torque if the stator voltage is reduced by 20%
b. The resistance that should be added to the rotor circuit to achieve the maximum torque at starting

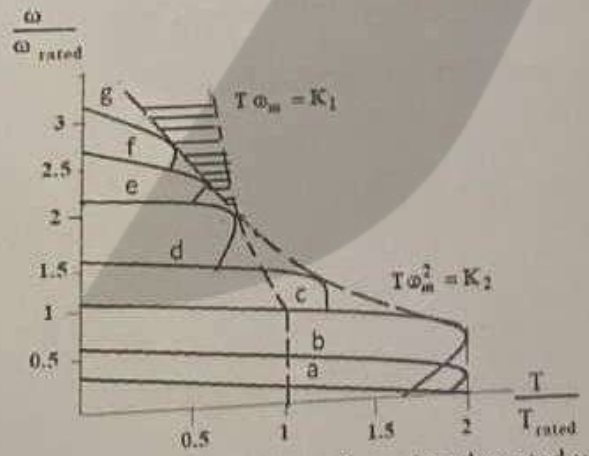
Solution

Electric Drive Final Exam 50 Point Instructor: Prof. Mahdi Alshamasin date:23-01-2021
Time of Exam: Two class hours Student Name:

Q1: (30 marks)
Choose the most appropriate answer for each item (a, b, c, or d) by filling in the right choice in the table given at the end of this question

1. Converter of one quadrant drive is
   a. three-phase semiconverter b. three-phase full-wave-converter
   c. three-phase half-wave-converter d. single-phase full-wave-converter
2. An 4-pole, 3-phase, 50 Hz induction motor is operating at a speed of 1450 rpm. What is the rotor frequency in Hz?
   a. 1.34Hz b. 1.67Hz c.40Hz d. none of these
3. An 4-pole, 3-phase, 50 Hz induction motor is operating at a speed of 1450 rpm. What is the stator field angular speed rad/s?
   a. 78.5 b. 314 c.157 d. none of these
4. In 120 degrees' voltage source inverter, the output line voltage is
   a. 0.5 of the dc link voltage value b. 0.707 of the dc link voltage value
   c. 0.816 of the dc link voltage value d.0.47 of the dc link voltage value
5. Control which can provide very accurate torque and speed control for industrial applications is
   a. open-loop scalar variable frequency drive (VFD) b. open-loop vector VFD
   c. closed-loop vector VFD d. closed-loop scalar VFD
6. Slip energy in the rotor circuit of induction motor is
   a. $T\omega_o$ b. $T\omega_o(1-s)$ c. $T\omega_o$ d. $T\omega_o s$
7. For solid-state control of the external resistance in induction motor rotor circuit, the effective resistance is
   a. $(1-\delta)R_{ex}$ b. δR_{ex} c. R_{ex} d. none of these
8. Speed is controlled in mechanical cascade of slip ring ac motors by changing
   a. ac motor rotor field frequency b. dc motor field voltage c. ac motor rotor losses d. load torque
9. Static Kramer drive system is suitable for
   a. drilling machine b. lathe c. escalator d. fan

- For the given characteristics, answer questions (10-14)



10. Current is constant and equal to the rated value for the following characteristics
    a. c-e b. e-g c. c-g d. a-e
11. Voltage is constant and equal to the rated value for the following characteristics
    a. a-g b. e-g c. c-g d. a-e
12. characteristics of flux weakening region are
    a. a-g b. c-g c. a-c d. a-e
13. characteristics of constant torque region are
    a. a-g b. c-g c. a-c d. a-e
14. Characteristics in the figure are suitable for the electric vehicle in
    a. city cruise b. high-way cruise c. no-load mode d. high-way and city cruise

Ans

Q2: (10 marks)
Draw mechanical characteristics for
a. Static Scherbuis Drive System
b. Static Kramer Drive System
c. Induction motor when using solid-state control of the rotor resistance circuit
d. squirrel-cage motor controlled by variable voltage variable frequency with constant load
E. squirrel-cage motor controlled by variable voltage constant frequency with fan load

a. Static Scherbuis Drive System

b. static Kramer Drive System

c. solid-state control of the rotor resistance circuit

d. variable voltage variable frequency with constant load

a.280V b.335V c. 400V d.none of these

28.For single-phase full-wave-converter drives, regenerative braking occurs in

a. 1st quadrant b.2nd quadrant c. 3rd quadrant d. 4th quadrant

29. α_{min} below which the thyristor cannot be triggered in single-phase half-wave-converter drives

a. $\alpha_{min} = \tan^{-1}\left(\frac{E_a}{V_m}\right)$ b. $\alpha_{min} = \pi - \sin^{-1}\left(\frac{E_a}{V_m}\right)$

c. $\alpha_{min} = \sin^{-1}\left(\frac{E_a}{V_m}\right)$ d. $\alpha_{min} = \cos^{-1}\left(\frac{E_a}{V_m}\right)$

30. Using three-phase full-wave semi converter, the armature voltage of high- loaded sep. excited dc motor is

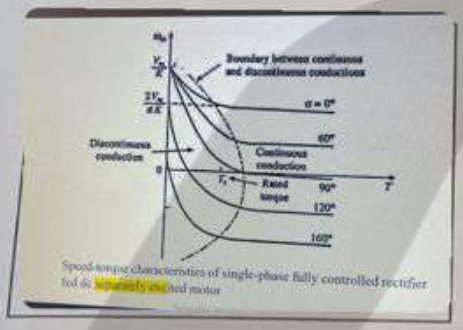
a. $\frac{3\sqrt{3}V_m}{\pi}\cos\alpha$ b. $\frac{3\sqrt{3}V_m}{2\pi}(1+\cos\alpha)$ c. $\frac{3\sqrt{3}V_m}{2\pi}\cos\alpha$ d. $2.34V_s\cos\alpha$

Q	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans																				
Q	21	22	23	24	25	26	27	28	29	30										
Ans																				

Q2: (10 marks)

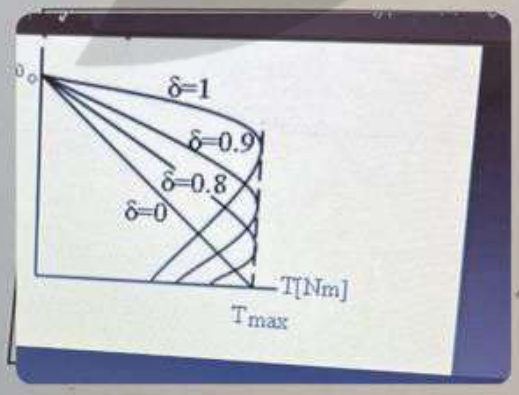
Draw mechanical characteristics for

a. sep. excited dc motor fed by controlled single-phase full-wave-Converter



Speed-torque characteristics of single-phase fully controlled rectifier fed dc separately excited motor

b. solid-state induction motor using a rotor chopper-controlled external resistor



c. slip losses control of wound rotor ac motors by mechanical cascade

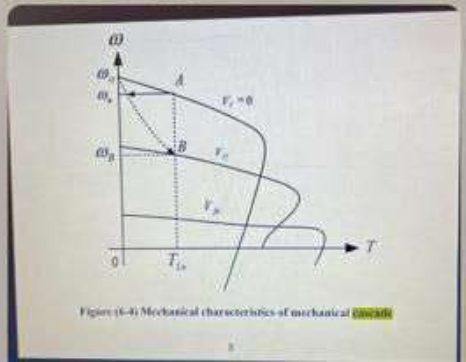


Figure (6-4) Mechanical characteristics of mechanical cascade

Al-Balqa' Applied University

Faculty Of Engineering Technology

Mechatronics Engineering Dept.

Time of Exam: 80 Min.

12/09/2022

Fundamentals of Elec. Drive

Final exam
50 Points

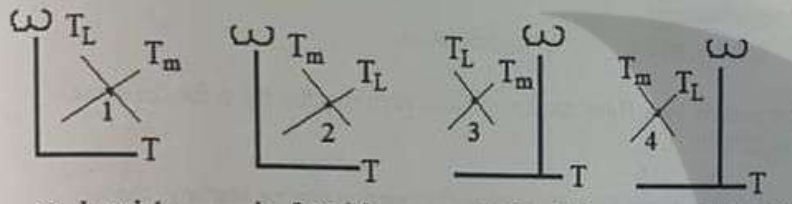
Prof. Dr. Mahdi Alshamasin	Instructor

الرقم الجامعي: الشعبة: اسم الطالب

Q1: (35 marks)

Answer the following questions by filling the table at the end of this question

1. Unstable points of equilibrium are



a. 1 and 4 b. 2 and 3 c. 1 and 3 d. 2 and 4

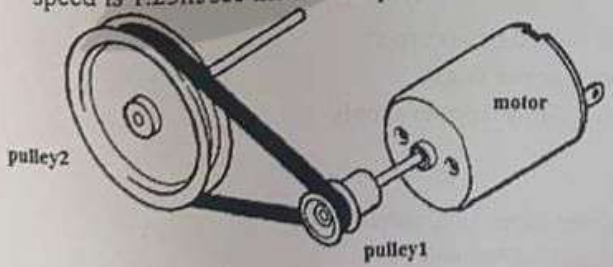
2. Conveyor torque –speed characteristic is:
   a. Constant torque characteristic b. Linearly rising torque characteristic
   c. Non-Linearly rising torque characteristic d. Non-Linearly decreasing torque characteristic
3. If the torque developed by the motor is $T = T_L \mp J\frac{d\omega}{dt}$, the motor is working at

a. no-load condition b. transient condition c. S.S.P d. none of these

4. The motion equation $T = T_L + J\frac{d\omega}{dt} + B\omega$ can be solved analytically if
   a. T and T_L are constants b. T or T_L is neither constant nor proportional to speed
   c. T and T_L are constants or proportional to speed d. none of these.
5. The equivalent moment of inertia referred to motor shaft calculated depending on:

a. Heating (electric) power conservation law b. Mechanical power conservation law
c. Electro-magnetic energy conservation law d. Kinetic energy conservation law

6. In the figure below, diameters of pulley 1 and pulley2 are 10cm and 50cm respectively, if the belt speed is 1.25m/sec the motor speed is



a.240rpm b. 250rpm c. 12r/sec d. none of these

7. One of the following dc motors does not operate in regenerating mode:

a. Parallel field dc motor b. Series field dc motor
c. Compound field dc motor d. Separately excited dc motor

8. When the motor speed is controlling by stator voltage changing, the permissible torque is governed by

a. $T_{per} = T_n S/S_n$ b. $T_{per} = T_n S_{sc}/S$ c. $T_{per} = T_n S_n/S$ d. $T_{per} = T_{cr}S_{cr}/S$

9. Component determines the motor starting time is
   a. motor inertia b. motor load c. speed change rate d. all of these

Drive Final Exam: 50 Points Instructor: Prof. Mahdi Ashtiyaniyan, date:12/01/2020
of Exam: Two (02) hours Student Name:

(30 marks)
mose the most appropriate answer for each item (a, b, c, or d) by filling in the right choice
ble given at the end of this question

1. Which of the following is a key component of an electric drive system?
a. electric motor b. power converter c. controller d. all of the above
2. What is the main function of the controller in an electric drive system?
a. to provide mechanical support b. to regulate voltage and current
b. to drive the load directly d. to store electrical energy
c. to convert the purpose of a power converter in an electric drive system?
d. to control the direction of the load
e. to convert and regulate electrical energy for the motor
f. to act as a load for the motor
4. Which parameter is controlled in a field-controlled DC motor drive?
a. armature current b. torque c. rotor speed d. field current
5. In ____ machines, a large torque of short duration is required during operation, otherwise, the torque nearly zero.
a. electric hammer b. cutting c. drilling d. lathe
6. The equivalent moment of inertia referred to motor shaft calculation is based on the
a. mechanical power conservation law b. kinetic energy conservation law
b. electro-magnetic energy conservation law d. heating (electric) power conservation law
c. electro-magnetic energy conservation law
7. Referring to the multi-quadrant motor operation, the correct statement is
a. the motor can provide motoring and braking operations for forward direction only
b. the motor can provide motoring and braking operations for both forward and reverse directions
c. the motor can provide forward braking when the motor overloaded
d. the motor can provide forward braking when the motor unloaded
8. An electric motor with a moment of inertia of 2 kg.m² is connected through a gearbox with a gear ratio of 10 to a drum with a moment of inertia of 8 kg.m². The total (equivalent) moment of inertia of the drive system is
a. 0.1 kg.m² b. 0.28 kg.m² c. 2.08 kg.m² d. 2.8 kg.m²
9. A crane motor lowers a load of 10 N.m through a gearbox with a gear ratio of 5. If the system efficiency is 0.8, the load torque referred to the motor shaft is
a. 1.8N.m b. 2.5N.m c. 40N.m d. 1.6N.m
10. The hardness of series field dc motors torque-speed characteristic is
a. infinity b. positive c. zero d. negative
11. Transient conditions of electric drive system are
a. starting, braking, speed change and speed reversal operations
b. starting, braking, steady-state and speed reversal operations
c. starting, braking, speed change and no-load mode operations
d. starting, braking, speed change and locked-rotor mode operations
12. Components determine the motor starting time are
a. motor size only b. motor mass, size and speed c. motor inertia only d. load inertia only
13. The suitable principle for automatic starting, braking and reversing of elevator is
a. speed principle b. time principle c. displacement (position) principle d. current principle
14. Allowed load torque of dc motor speed control method depends on the variation of
a. speed direction b. motor speed c. armature voltage d. motor flux
15. The following profile refers to
a. duty type S3 b. duty type S5 c. duty type S6 d. duty type S

22. If mechanical time-constant related to electro-magnetic time-constant of drive system value is less than four, then drive system characteristic equation has:

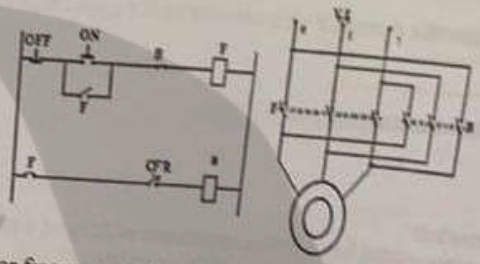
a. Two different real roots
b. Two equal real roots
c. Two different complex roots
d. Two equal complex roots

23. Centrifugal relay is used for automatic starting and braking of electric motors according to:

a. Position principle
b. Voltage principle
c. Speed Principle
d. Time principle

24. The given electrical diagram shows

a. direct starting and mechanical braking
b. Starting in forward direction and plugging braking
c. Starting in forward direction and dynamic braking
d. Forward working and backward working



25. In speed regulation of three-phase ac motors by stator frequency variation with nominal voltage, the frequency should not be less than the nominal value to protect the motor from

a. Overheating
b. Under loading
c. Over voltage
d. Over loading

26. According to the AC motor speed control by varying stator voltage with fixed frequency, Load torque should be subjected to have constant

a. Load torque
b. Critical torque
c. Starting torque
d. Motor loading capacity

27. Increasing the input supply frequency of three-phase ac motor more than nominal frequency with fixed stator voltage will lead to

a. Decrease stator field angular speed
b. Increase motor starting torque
c. Decrease field flux
d. Increase motor load torque

28. The parallel connection of ac motor stator windings will cause variation of:

a. Stator voltage
b. No-load speed
c. Load torque
d. Stator frequency

Q	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Ans														

Q	15	16	17	18	19	20	21	22	23	24	25	26	27	28
Ans														

Q2: Given the circuit diagrams of control and power for squirrel-cage motor below (8 marks)

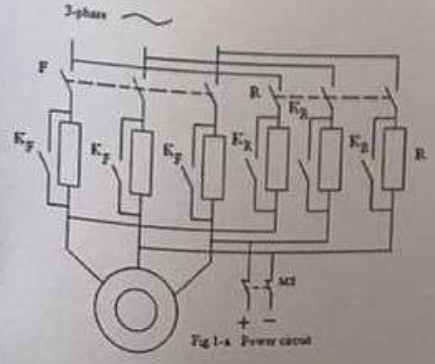


Fig 1-a Power circuit

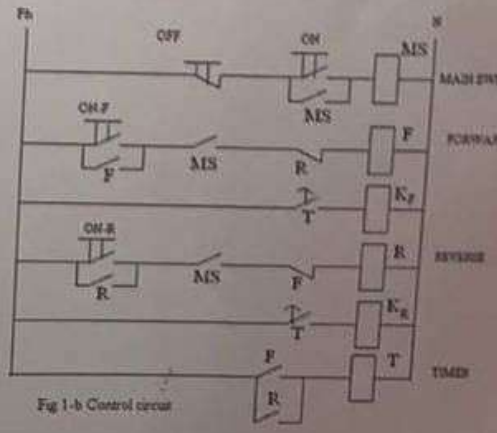


Fig 1-b Control circuit

a. Draw the static mechanical characteristics

b. Draw the dynamic characteristics for speed and Torque

Q3: (7 marks)

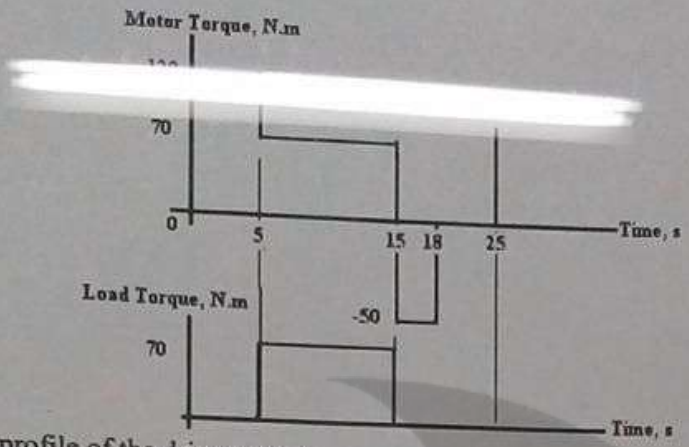
An induction motor has a stator resistance of 3Ω and a rotor resistance referred to the stator is 2Ω, the critical inductive reactance X_{cr}=10Ω, calculate

a. The change in starting torque if the stator voltage is reduced by 20%
b. The resistance that should be added to the rotor circuit to achieve the maximum torque at starting

Solution

Question 3 **(10 points)**

A dive system has the following motor torque and load torque profiles. The total (equivalent) moment of inertia of the system is 1.2 kg.m^2.



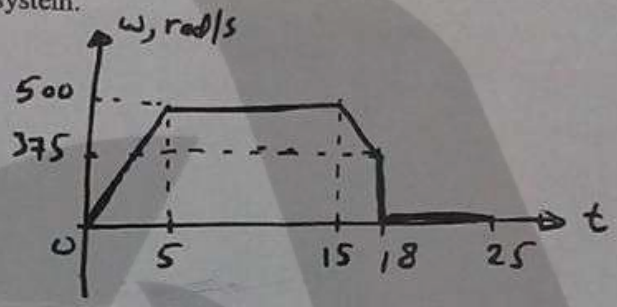
(1) Plot the speed profile of the drive system.

Answer:

$$\frac{d\omega}{dt} = \frac{T - T_L}{J_e}$$

$0 \le t \le 5$; $\frac{d\omega}{dt} = 100$ rad/s^2

$15 \le t \le 18$; $\frac{d\omega}{dt} = -41.66$ rad/s^2



4

(2) What is the type (class) of duty cycle?

Answer:

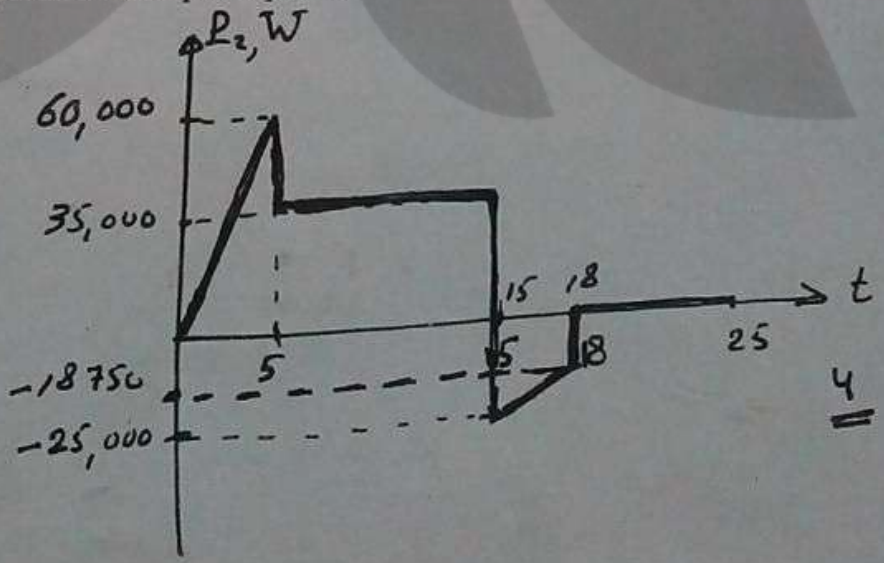
Intermittent Periodic with starting and braking.

2

(3) Plot the developed output mechanical power profile.

Answer:

$P_2 = \omega \cdot T$



4

7 **The phase uncontrolled rectifiers used in speed control of dc motors converts fixed ac supply voltage into.....output voltage.**

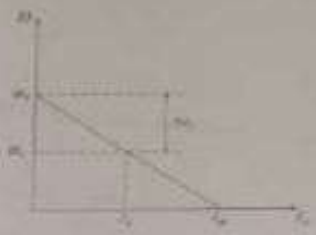
a. fixed dc b. variable dc c. variable frequency d. full rectified ac

8 **Speed control method which is used for transport at high-way roads (out of city) is**

a. Changing the stator voltage of the induction motor
b. Inserting resistance into the rotor circuit of induction motor
c. Using of VVVF method for all $V < V_n$
d. Changing the frequency of induction motor while keeping its stator voltage constant

9 **In the next figure, the motor works in**

a. Motoring mode
b. Forward braking mode
c. Reverse braking mode
d. Plugging mode

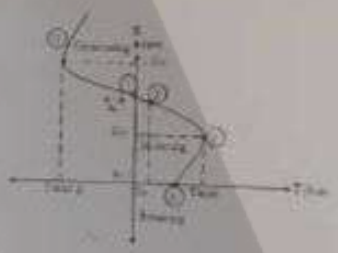


10 **Speed control method which is used for cranes is**

a. Changing the stator voltage of the induction motor
b. Inserting resistance into the stator circuit of induction motor
c. Using of VVVF method for all $f > f_n$
d. Inserting resistance into the rotor circuit of induction motor

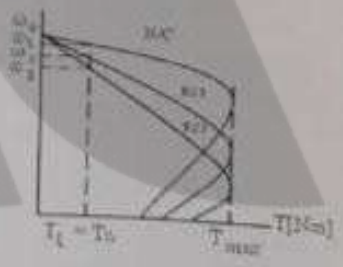
11 **Point 5 in the next figure represents:**

a. maximum torque on motor operation
b. maximum torque on generating mode
c. maximum torque on generating mode
d. synchronous speed



12 **For the figure below, the correct phrase is**

a. Speed range is high
b. Hardness is high
c. Method is economic
d. The method is used for transporting machineries



13 **The single- phase rectifier used in voltage source inverter to control the speed of ac motor of power.**

a. 5kw b. < 5kw c. > 5kw d. < 10kw

14 **Dynamic braking of separately induction motor is achieved by**

a. disconnecting the power supply and feeding two stator phases with dc voltage
b. disconnecting the power supply and shorten two stator phases with pure resistance
c. reversing the motor
d. Increasing the load to be more than the starting torque
e.